MATH 261, Homework Set 9 Name: _____

1. Let S be the outward oriented sphere $x^2 + y^2 + z^2 = 1$, and let $\mathbf{F} = z\mathbf{i} - 5\mathbf{k}$ be a vector field. Set up, but do NOT solve, the flux integral $\iint_S \mathbf{F} \cdot \mathbf{N}\, d\sigma$

Final answer (integral setup)

2. Let S be the surface obtained by rotating the curve $y = x^3$ for $1 \leq x \leq 3$ around the x-axis, oriented so the normal at the point $(2, 8, 0)$ is pointing in the positive y direction. Let $\mathbf{F} = y\mathbf{i} + x\mathbf{j} + z\mathbf{k}$ be a vector field. Set up, but do NOT solve, the flux integral $\iint_S \mathbf{F} \cdot \mathbf{N}\, d\sigma$

Final answer (integral setup)

3. Let S be the surface given by the portion of the sphere $x^2+y^2+z^2=10$ which is above the plane $z=-1$ and below the plane $z=1$ oriented so the normal always points away from the origin. Let $\mathbf{F}=1\mathbf{i}+3\mathbf{j}+2\mathbf{k}$ be a vector field. Set up, but do NOT solve, the flux integral $\iint_S \mathbf{F}\cdot\mathbf{N}\,d\sigma$

Final answer (integral setup)

4. Let S be the surface given by the portion of the plane $2x+3y+z=6$ which lies in the first octant, oriented so that the normal always points in the positive z direction. Let $\mathbf{F}=\langle 2,1,-2\rangle$ be a vector field. Compute the flux integral $\iint_S \mathbf{F}\cdot\mathbf{N}\,d\sigma$

Final answer

5. Let S be the surface given by the portion of the plane $2x + 3y + z = 6$ which lies in the first octant, oriented so that the normal always points in the positive z direction. Let $\mathbf{F} = \langle 1, 0, -2 \rangle$ be a vector field. Compute the flux integral $\iint_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$

Final answer

6. Let S be the surface given by the portion of the cone $z = \sqrt{x^2 + y^2}$ which is above the plane $z = 1$ and below the plane $z = 2$ oriented so that the normal always points in the negative z direction. Let $\mathbf{F} = \langle 1, 1, z^2 \rangle$ be a vector field. Compute the flux integral $\iint_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$

Final answer

7. Let S be the outward oriented surface consisting of the upper half of the sphere with radius 1, $x^2 + y^2 + z^2 = 1$ for $z \geq 0$, and the disc $x^2 + y^2 \leq 1$. Let $\mathbf{F} = \langle y, -x, 1 \rangle$ be a vector field. Compute the flux integral $\iint_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$

Final answer

8. Let S be the outward oriented surface consisting of the upper half of the sphere with radius 1, $x^2 + y^2 + z^2 = 1$ for $z \geq 0$, and the disc $x^2 + y^2 \leq 1$. Let $\mathbf{F} = \langle 0, 0, z \rangle$ be a vector field. Compute the flux integral $\iint_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$

Final answer

9. Let S be the outward oriented surface consisting of the portion of the cylinder $x^2+y^2 = 1$ below the plane $z = 1$ and above the plane $z = -1$, as well as the portions of planes $z = 1$ and $z = -1$ which are within the cylinder (so the cylinder has a lid on both ends). Let $\mathbf{F} = \langle x, y, z^2 \rangle$ be a vector field. Compute the flux integral $\iint_S \mathbf{F} \cdot \mathbf{N}\, d\sigma$

Final answer

10. Let $\mathbf{F} = \langle y^2+z^2, x^2+z^2, x^2+y^2 \rangle$ be a vector field, and let C be the square bounded by the lines $x = \pm 1$, $y = \pm 1$, and $z = 0$, traversed counterclockwise. Use Stokes' theorem to compute the line integral $\int_C \mathbf{F} \cdot \mathbf{T}\, ds$

Final answer

11. Let $\mathbf{F} = \langle x^2, 2x, z^2 \rangle$ be a vector field, and let C be the ellipse $4x^2 + y^2 = 4$, and $z = 0$, traversed counterclockwise. Use Stokes' theorem to compute the line integral $\int_C \mathbf{F} \cdot \mathbf{T}\, ds$

Final answer

12. Let S be the outward oriented surface consisting of the top half of the sphere $x^2 + y^2 + z^2 = 16$ and the disc $x^2 + y^2 \leq 16$ at height $z = 0$. Let $\mathbf{F} = \langle x^2, z^2 y, z y^2 \rangle$ be a vector field. Use Stokes' theorem to compute $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N}\, d\sigma$

Final answer

13. Let S be the upper half of the sphere $x^2 + y^2 + z^2 = 1$ for $z > 0$, oriented upward. Let $\mathbf{F} = \langle y, z, x \rangle$ be a vector field. Use Stokes' theorem to compute $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma$

Final answer

14. Let S be downward oriented surface consisting of the portion of the sphere $x^2+y^2+z^2 = 25$ which lies below the plane $z = -3$. Let $\mathbf{F} = \langle zy, -zx, e^{z^2} \rangle$ be a vector field. Use Stokes' theorem to compute $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma$

Final answer

15. Let S be the outward oriented sphere $x^2 + y^2 + z^2 = 1$. Let $\mathbf{F} = \langle z^2, 0, -xy \rangle$ be a vector field. Use Stokes' theorem to compute $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma$

Final answer

16. Let S be the downward oriented surface given by the portion of the cone $z = \sqrt{x^2 + y^2}$ which is above the plane $z = 1$ and below the plane $z = 2$. Let $\mathbf{F} = \langle yz, 0, z^2 \rangle$ be a vector field. Use Stokes' theorem to compute $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma$

Final answer

17. Let S be the outward oriented surface consisting of the portion of the cylinder $x^2+y^2 = 1$ below the plane $z = 1$ and above the plane $z = -1$. Let $\mathbf{F} = \langle y^2, x, 1\rangle$ be a vector field. Use Stokes' theorem to compute $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N}\, d\sigma$

Final answer

18. Let S be the outward oriented surface consisting of the portion of the graph $z = 10 - x^2 - y^2$ above the plane $z = 1$. Let $\mathbf{F} = \langle y, 0, x\rangle$ be a vector field. Use Stokes' theorem to compute $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N}\, d\sigma$

Final answer

19. (Conceptual problems) Let f be a scalar function defined on $\mathbb{R}^3$ and let $\mathbf{F}$ be a vector field defined on $\mathbb{R}^3$. Assume that f and all components of vector field $\mathbf{F}$ have continuous second partial derivatives. For each of the following expression, state (1) whether each expression makes sense mathematically. If it does, state (2) whether you get a scalar or vector field; and (3) whether it is always 0 or $\mathbf{0}$ if the object that it gives you is a vector.

(a) $\nabla \times (\nabla \times \mathbf{F})$

Makes sense?	**Scalar / vector?**	**Always zero?**

(b) $\nabla \cdot (\nabla \cdot \mathbf{F})$

Makes sense?	**Scalar / vector?**	**Always zero?**

(c) $\nabla \times (\nabla \cdot \mathbf{F})$

Makes sense?	**Scalar / vector?**	**Always zero?**

(d) $\nabla \times (\nabla f)$

Makes sense?	**Scalar / vector?**	**Always zero?**

(e) $\nabla(\nabla \cdot \mathbf{F})$

Makes sense?	**Scalar / vector?**	**Always zero?**

(f) $\nabla \cdot (\nabla f)$

Makes sense?	**Scalar / vector?**	**Always zero?**

(g) $\nabla \cdot (\nabla \times \mathbf{F})$

Makes sense?	**Scalar / vector?**	**Always zero?**